

VIKAS SINGH

Data + AI Engineer | AI & LLM Applications | Snowflake, RAG & Agentic Systems

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience building production data platforms and AI-powered applications on top of them. Led an enterprise-scale Redshift-to-Snowflake migration (25,000+ SQL queries, 8,000+ objects, 5B+ row datasets), and independently designed an Agentic RAG system (LangGraph, Vector DB, FastAPI) combining LLM orchestration, vector search, and automated validation. Skilled across the data-to-AI stack — ETL/ELT, cloud data warehousing (Snowflake, Redshift, AWS), and applied GenAI (RAG, LLM orchestration, agent design) — with a track record of cutting query runtime by 70%+ and automating 30+ engineering hours/week of manual work.

EXPERIENCE

Associate Consultant – Data Engineering, Hakkoda (IBM)

Dec 2024 – Present

- **Led enterprise migration** of analytical workloads from Amazon Redshift and MySQL to Snowflake, converting 25,000+ SQL queries and 8,000+ database objects across datasets up to 5B+ rows — delivered on schedule with no disruption to downstream reporting.
- **Optimized Snowflake SQL workloads and warehouse configurations**, reducing aggregation runtime from 9 hours to 2.5 hours (a 72% improvement), directly cutting compute costs and unblocking faster downstream analytics.
- **Designed and built scalable ETL/ELT pipelines** using Snowflake, Python, Java, and AWS to support reliable, repeatable data delivery across the migration program.
- **Built an enterprise data validation framework** (schema, row-hash, statistical, query-level, and custom SQL checks) that became the standard QA gate for migration sign-off, catching discrepancies before they reached production.
- **Engineered a configurable Patch Installer** supporting one-to-one, one-to-many, and many-to-many deployments with rollback and dependency validation, reducing manual deployment errors.
- **Automated deployment workflows**, saving 30+ engineering hours per week and improving platform stability by 45–50%.
- **Partnered with a cross-region, distributed engineering team** to align on requirements and coordinate delivery across time zones.

PROJECTS

HR AI Chatbot — Agentic RAG

Personal Project · LangGraph, FastAPI, Vector DB, Groq/Anthropic, Tavily

- **Designed and built an end-to-end Agentic RAG application:** an India-centric HR assistant orchestrated by a LangGraph state machine that routes, retrieves, grades, and rewrites queries across a private knowledge base and live web search.
- **Built the retrieval layer** on a Vector DB (cosine similarity) with HuggingFace embeddings (all-MiniLM-L6-v2), ingesting PDF/DOCX/MD/TXT via a custom chunking pipeline.
- **Implemented an 8-node agentic workflow** (route → retrieve → grade → generate/rewrite/fallback) with automatic query rewriting and web-search fallback (Tavily) when KB results were graded insufficient.
- **Integrated dual LLM providers** (Groq and Anthropic Claude) behind a single interface with context-aware, source-cited answer generation.
- **Shipped a full-stack production app:** FastAPI backend (/chat, /ingest, /kb/stats), admin console for ingestion, chat UI with citations/trace view, and DuckDB audit logging.

DataValidator Pro

Personal / Team Project

- **Built an automated data validation platform** (Streamlit UI + CLI) for Redshift-to-Snowflake migrations, replacing manual, spreadsheet-based checks.
- **Implemented schema, row-hash, query, statistical, duplicate, and custom SQL validation**, with automated Excel reconciliation reporting for stakeholder sign-off.
- **Integrated validation checks into the CI/CD pipeline**, catching data discrepancies before production and reducing rollbacks.

TECHNICAL SKILLS

AI / LLM & GenAI: Agentic RAG, LangGraph, LLM Orchestration (Groq, Claude), Prompt Engineering, Vector Search (Vector DB), HuggingFace Embeddings

Languages: Python, SQL, Java, C++

Data Warehousing: Snowflake, Amazon Redshift, MySQL

Data Engineering: ETL/ELT, Data Migration, Data Validation, Data Modeling, SQL Optimization, CI/CD

Big Data & Orchestration: Apache Spark, Apache Airflow, Docker

Cloud (AWS): S3, CloudWatch, CodePipeline, CodeBuild, DynamoDB, Secrets Manager

App / API & Tools: FastAPI, Streamlit, REST APIs, Git, GitHub, Bitbucket, dbt, Jira, Confluence, Excel, DuckDB

EDUCATION

B.Tech in Computer Science (Artificial Intelligence)

2021 – 2025

Swami Keshvanand Institute of Technology

CERTIFICATIONS

- SnowPro Core Certified — Snowflake
- Databricks Certified Associate Developer for Apache Spark – Python — Databricks